

PRIVACY-PRESERVING PUBLIC TRANSPORT DEMAND FORECASTING SYSTEM USING FEDERATED LEARNING

A PROJECT REPORT

Submitted by

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CERTIFICATE

This is to certify that the report entitled **Privacy-Preserving Public Transport Demand Forecasting System Using Federated Learning** submitted by **AASIYA M (TKM24MCA-2001)** to TKM College of Engineering affiliated to APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Master of Computer Application is a bonafide record of the project work carried out by her under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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DECLARATION

I undersigned hereby declare that the project report **Privacy-Preserving Public Transport Demand Forecasting System Using Federated Learning** submitted for partial fulfillment of the requirements for the award of degree of Master of Computer Application of the APJ Abdul Kalam Technological University, Kerala, is a bonafide work done by me under supervision of **Prof. Natheera Beevi M.** This submission represents my ideas in my own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

Place: Kollam

AASIYA M

Date: 19/03/2026

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ABSTRACT

Public transport systems require accurate demand forecasting to support efficient scheduling, fleet management, and service planning. However, ridership data is often distributed across multiple transport operators and cannot be centrally shared due to privacy concerns, regulatory restrictions, and data ownership constraints. This limits the effectiveness of traditional centralized forecasting approaches.

This project presents a privacy-preserving public transport demand forecasting system using Federated Learning (FL). In this approach, multiple transport operators collaboratively train a global model without sharing raw data. Each participant performs local training on its own dataset, and only model parameters are shared with a central coordinator for aggregation. The global model is updated iteratively using federated optimization techniques to improve prediction performance.

The system is implemented using real-world public transport data, covering multiple routes and long-term historical demand patterns. A machine learning model based on Stochastic Gradient Descent is used to enable efficient and scalable federated training. The proposed system also supports dynamic participation of operators and provides analytical insights such as demand trends and anomaly detection.

The results demonstrate that federated learning can effectively enable collaborative demand forecasting while preserving data privacy. The system provides a practical and scalable solution for real-world public transport networks where data sharing is restricted.

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Chapter 1

INTRODUCTION

Public transport systems are a fundamental component of modern urban infrastructure, enabling the efficient movement of people across cities. With rapid urbanization and increasing population density, the demand for reliable and efficient public transport services has grown significantly. Accurate demand forecasting plays a crucial role in optimizing scheduling, improving service reliability, and ensuring efficient fleet allocation.

However, predicting transport demand in real-world scenarios is challenging due to the fragmented nature of data. Transport data is often distributed across multiple independent operators or administrative regions, making centralized data collection difficult. Privacy concerns, regulatory restrictions, and commercial sensitivity further limit the sharing of such data, reducing the effectiveness of traditional forecasting approaches.

Traditionally, demand forecasting relies on centralized systems where data from multiple operators is aggregated and processed in a single repository. While these systems can achieve high accuracy, they assume full data availability and uniform data distribution. In practice, transport data exhibits significant heterogeneity (non-IID characteristics), where demand patterns vary across regions, routes, and time periods. This limits the applicability of conventional machine learning models in real-world deployments.

The Chicago Transit Authority (CTA) dataset provides a comprehensive longitudinal record of 123 bus routes, spanning from January 2001 to November 2025. It includes monthly ridership, average weekday and weekend demand, and geographic information of routes. This dataset enables realistic demand forecasting while capturing long-term temporal patterns and spatial variability across different

operational zones.

To address the challenges of data privacy and decentralization, Federated Learning (FL) has emerged as a promising solution. In federated learning, multiple participants collaboratively train a shared global model without exchanging raw data. Each operator trains a local model using its own dataset and shares only model parameters with a central coordinator. The coordinator aggregates these updates using federated optimization algorithms such as Federated Averaging (FedAvg) and Federated Proximal (FedProx), iteratively improving the global model while preserving data privacy.

In this project, a privacy-preserving public transport demand forecasting system is developed using Federated Learning. The CTA dataset is partitioned into eight geographically coherent clients using K-Means clustering on route coordinates, simulating realistic multi-operator environments. Each client trains a local model using Stochastic Gradient Descent (SGDRegressor), enabling efficient iterative updates and convergence across federated training rounds.

The system is designed to support dynamic onboarding, allowing new operators to join the federation at any stage without disrupting ongoing training. This ensures scalability and adaptability in real-world deployments where participating entities may change over time.

Beyond prediction, the proposed system integrates an insights engine to provide actionable analytics. It performs year-on-year trend analysis to identify growth and decline patterns, Z-score-based anomaly detection to detect sudden demand fluctuations, and service pattern classification to distinguish between commuter-dominant and leisure-oriented routes. These insights support data-driven decision-making for transport authorities.

Overall, the proposed system combines privacy preservation, decentralized learning, and analytical intelligence into a unified framework. By enabling collaborative forecasting without data sharing, it provides a practical and scalable solution for real-world public transport systems.

1.1 Existing Systems

Traditional public transport demand forecasting systems primarily rely on centralized architectures, where data from multiple transport operators is collected and processed in a single repository. Statistical models and machine learning techniques are then applied to generate demand predictions. While such systems perform well when complete data is accessible, they are impractical in real-world scenarios where data sharing is restricted due to privacy concerns, regulatory policies, and commercial sensitivity.

Centralized systems introduce several limitations, including data governance challenges, increased security risks, and a single point of failure. Additionally, these systems assume uniform data distributions, which is rarely the case in real-world transport networks where demand varies significantly across regions, resulting in non-IID data characteristics.

Recent research has explored federated learning for transportation applications such as ride-hailing demand prediction and passenger flow estimation. These approaches enable decentralized training while preserving data privacy. However, most existing works remain theoretical or experimental and do not provide complete system-level implementations.

Furthermore, current federated learning solutions often lack essential features required for practical deployment, such as dynamic client onboarding, robustness to heterogeneous data, scalability, and intuitive monitoring interfaces. Many systems also focus only on prediction accuracy without integrating analytical insights for operational decision-making.

1.2 Problem Statement

Accurate demand forecasting in public transport systems requires comprehensive datasets spanning multiple operators and geographic regions. However, such data is inherently distributed and cannot be centralized due to privacy concerns, regulatory constraints, and commercial sensitivity. This limitation prevents

traditional models from fully utilizing available data, resulting in reduced prediction accuracy and limited collaboration among operators.

In addition, transport data exhibits heterogeneous and non-IID characteristics, where demand patterns differ significantly based on location, time, and user behavior. Conventional machine learning models struggle to effectively handle such variability, leading to suboptimal performance.

Although federated learning provides a solution for decentralized training, most existing implementations do not address practical challenges such as dynamic onboarding of clients, scalability, robustness to non-IID data, and integration of actionable insights.

Therefore, there is a need for a scalable, privacy-preserving system that enables collaborative demand forecasting across multiple operators while effectively handling heterogeneous data and providing meaningful analytical insights.

1.3 Proposed System

The proposed system is a privacy-preserving public transport demand forecasting platform based on Federated Learning (FL). It enables multiple transport operators to collaboratively train a global model without sharing raw data, ensuring data privacy and security.

Each operator functions as a federated client, training a local machine learning model using its own historical data. The trained model parameters are periodically shared with a central coordinator, which aggregates them using federated optimization algorithms such as FedAvg and FedProx. While FedAvg performs weighted averaging of client updates, FedProx introduces a proximal constraint to improve convergence under non-IID data conditions.

The system utilizes SGDRegressor for local model training due to its efficiency and support for iterative updates. Each training round involves initializing local models with global parameters, performing local updates, and returning optimized parameters to the server.

To simulate real-world deployment, the dataset is partitioned into geograph-

ically coherent client groups, representing independent transport operators. The system also supports dynamic onboarding and removal of clients, ensuring flexibility and scalability.

An integrated insights engine enhances the system by providing:

- Trend analysis to identify long-term demand patterns
- Anomaly detection to detect sudden changes in ridership
- Service pattern classification based on weekday and weekend demand

A web-based interface is developed to visualize model performance, training progress, and analytical insights, enabling effective decision-making for transport authorities.

1.4 Objectives

The main objectives of this project are as follows:

- To design a privacy-preserving public transport demand forecasting system.
- To enable collaborative model training across multiple decentralized transport operators.
- To implement federated learning using FedAvg and FedProx without sharing raw data.
- To handle non-IID data distributions across different clients.
- To support dynamic onboarding and removal of participating operators.
- To ensure scalable and efficient model training across multiple federated clients.
- To validate prediction performance using a real-world longitudinal CTA dataset.
- To develop an analytical module for trend analysis, anomaly detection, and service pattern classification.

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- To provide a user-friendly web interface for monitoring, decision support, and actionable insights.

Chapter 2

LITERATURE SURVEY

A literature survey is a systematic review and critical analysis of previously published research, methodologies, and technologies related to a specific problem domain. It plays an important role in understanding existing approaches, identifying research trends, and recognizing limitations in current systems.

In the domain of public transport demand forecasting, significant research has been carried out using statistical models, machine learning techniques, and deep learning approaches. Traditional methods such as time-series forecasting models focus on capturing temporal patterns like trends and seasonality. More recent approaches utilize neural networks and hybrid models to improve prediction accuracy by learning complex spatial and temporal dependencies.

However, most existing demand forecasting systems rely on centralized data collection, where data from multiple transport operators is aggregated into a single system. Although this improves model performance, it introduces major challenges such as data privacy concerns, regulatory restrictions, and lack of inter-operator collaboration.

To address these issues, federated learning has emerged as a promising paradigm that enables decentralized model training without sharing raw data. In this approach, multiple clients collaboratively train a global model by sharing only model parameters. This ensures data privacy while still benefiting from distributed data sources.

Despite these advancements, many existing works focus primarily on algorithm design and experimental evaluation, with limited emphasis on building complete, real-world deployable systems. Challenges such as non-IID data handling, dynamic client participation, and integration of decision-support analytics remain

insufficiently explored.

This chapter reviews important research contributions in federated learning, time-series forecasting, and transportation analytics. It also highlights key limitations in existing approaches, which form the basis for the proposed system.

2.1 Purpose of Literature Survey

The purpose of this literature survey is as follows:

1. To study existing approaches for public transport demand forecasting.
2. To analyze machine learning and time-series models used for ridership prediction.
3. To understand federated learning techniques for decentralized model training.
4. To examine challenges such as data privacy, non-IID distributions, and system scalability.
5. To identify limitations and research gaps in existing systems.
6. To provide a strong foundation for the proposed privacy-preserving forecasting system.

2.2 Related Works

McMahan et al. (2017) [1] introduced the Federated Averaging (FedAvg) algorithm, which enables decentralized training by aggregating locally trained model parameters from multiple clients. Their work demonstrated that federated learning can significantly reduce communication overhead while preserving data privacy.

Li et al. (2020) [2] proposed the FedProx algorithm to address challenges arising from heterogeneous (non-IID) data distributions. By introducing a proximal term in the objective function, the method improves convergence stability in distributed environments.

Zhao et al. (2018) [3] analyzed the impact of non-IID data on federated learning performance and showed that data heterogeneity can significantly degrade model accuracy and convergence speed.

Box and Jenkins (1970) [4] developed classical time-series forecasting models such as ARIMA, which are widely used for demand prediction due to their ability to capture temporal patterns.

Ma et al. (2019) [5] proposed a deep learning-based approach using convolutional and bidirectional LSTM networks for metro ridership prediction, demonstrating improved accuracy in capturing spatial-temporal dependencies.

Liu et al. (2020) [6] applied federated learning to traffic flow prediction and showed that distributed models can achieve performance comparable to centralized systems while preserving privacy.

Chen et al. (2021) [7] extended federated learning to smart energy forecasting, demonstrating its applicability in real-world distributed systems.

Pedregosa et al. (2011) [8] introduced the Scikit-learn library, which provides scalable implementations of machine learning algorithms such as SGDRegressor, widely used for regression tasks.

Kairouz et al. (2021) [9] presented a comprehensive survey on federated learning, highlighting challenges such as communication efficiency, system heterogeneity, and privacy preservation.

Chicago Transit Authority (2025) [10] provides a large-scale public dataset of bus ridership, widely used for transportation analytics and benchmarking forecasting models.

From these studies, it is evident that while federated learning offers a privacy-preserving alternative to centralized systems, most existing works focus on algorithms rather than complete deployable frameworks.

2.3 Research Gap

From the literature, several limitations and research gaps are identified:

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- Most demand forecasting systems rely on centralized data collection, which is not feasible in multi-operator environments due to privacy and regulatory constraints.
- Existing federated learning research focuses mainly on algorithm development, with limited work on complete system-level implementations.
- Handling of non-IID data across clients is often discussed theoretically but rarely validated using real-world datasets.
- Dynamic client management, including onboarding and removal of participants, is not adequately addressed.
- There is a lack of integrated systems combining federated learning with analytics such as trend analysis, anomaly detection, and visualization.
- Most prior studies use short-term or simulated datasets instead of long-term real-world transport data.

The proposed system addresses these gaps by developing a complete federated learning-based demand forecasting framework using real-world data. It incorporates non-IID client simulation, dynamic operator onboarding, and an integrated insights engine, making it suitable for practical deployment in privacy-sensitive transport environments.

Chapter 3

METHODOLOGY

This chapter presents the complete methodology of the proposed Privacy-Preserving Public Transport Demand Forecasting System using Federated Learning (FL). The system is designed as an end-to-end pipeline that integrates data preprocessing, decentralized model training, and an insights engine for decision support.

Unlike traditional centralized approaches, the proposed system ensures that raw data remains within each transport operator's environment. Only model parameters are exchanged, enabling collaborative learning while preserving privacy. The methodology combines federated optimization, geographic client simulation, and analytical processing to deliver accurate predictions and actionable insights.

3.1 System Overview

The proposed system follows a decentralized federated learning architecture where multiple transport operators collaboratively train a global demand forecasting model. Each operator independently trains a local model using its private dataset.

The training process occurs in iterative communication rounds. In each round:

- Clients train local models on private data
- Model parameters are shared with the central server
- The server aggregates updates to form a global model
- The updated model is redistributed to clients

This process continues until convergence is achieved, ensuring improved model accuracy without compromising data privacy.

3.2 System Architecture

The overall architecture of the proposed system is illustrated in Figure 3.1. It represents the interaction between distributed clients, the federated server, and the analytics layer.

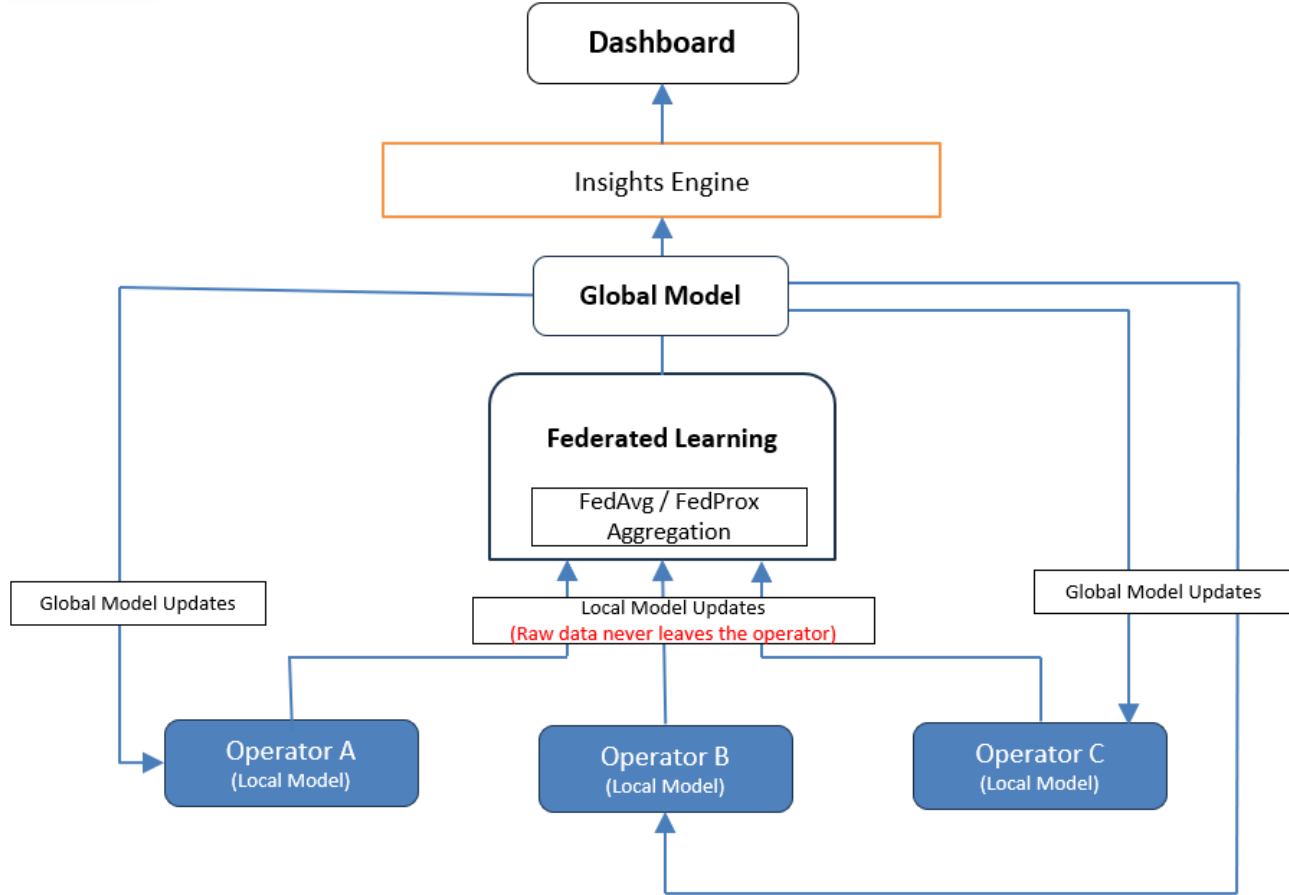


Figure 3.1: System Architecture of the Federated Learning-Based Transport Forecasting System

The architecture consists of the following components:

- **Operator Clients:** Each client maintains local ridership data and performs model training. Raw data is never shared outside the client environment.
- **Federated Learning Server:** Coordinates the training process by aggregating model parameters using FedAvg or FedProx algorithms.

- **Global Model:** Represents the collective knowledge learned from all participating clients and is iteratively refined.
- **Model Update Flow:** Only model parameters (coefficients and intercepts) are exchanged, ensuring privacy preservation.
- **Insights Engine:** Processes predictions into meaningful analytics such as trends and anomalies.
- **Dashboard Interface:** Provides visualization and monitoring tools for stakeholders.

This architecture ensures scalability, privacy, and adaptability in real-world multi-operator transport systems.

3.3 Methodology Workflow

The complete workflow of the proposed system is shown in Figure 3.2. It illustrates the sequential stages from data preprocessing to insight generation.

The workflow consists of the following stages:

1. Data Collection and Preprocessing
2. Feature Engineering
3. Geographic Client Simulation
4. Local Model Training
5. Federated Aggregation
6. Insight Generation

Each stage contributes to building a complete and deployable federated learning system.

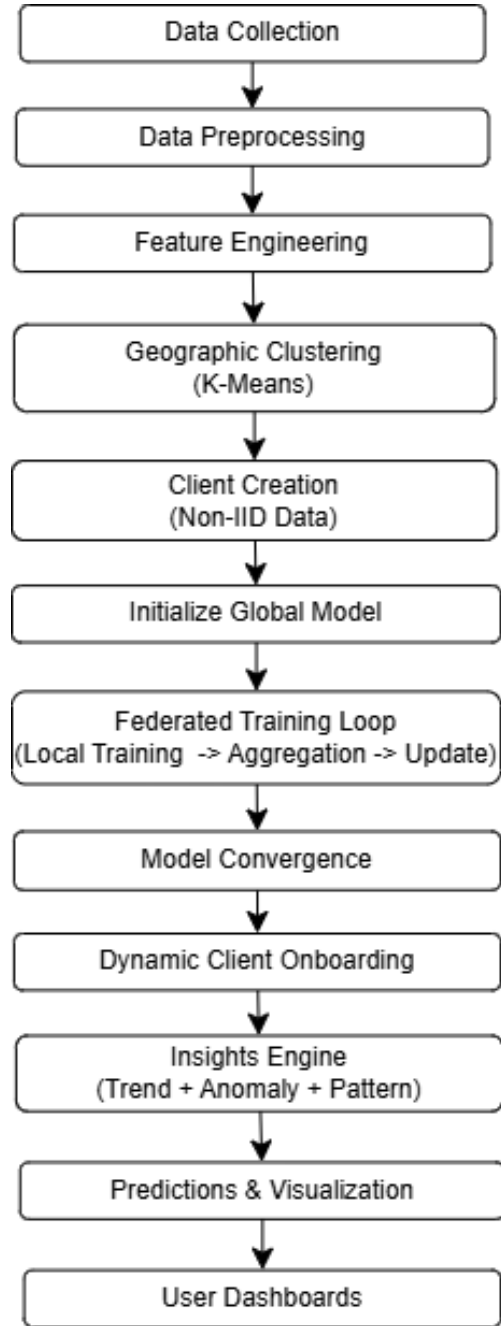


Figure 3.2: Overall Methodology Workflow of the Proposed System

3.4 Dataset Collection

The dataset used in this project is obtained from the Chicago Transit Authority (CTA) Open Data Portal. It includes monthly ridership data for 123 bus routes from January 2001 to November 2025.

This long-term dataset enables the model to capture seasonal patterns, de-

mand variability, and long-term trends.

3.5 Data Preprocessing

Data preprocessing ensures consistency and quality of the dataset before training.

- Temporal consistency validation across 24 years
- Removal of missing and inconsistent records
- Retention of 34,749 clean data points
- StandardScaler normalization applied per client
- Scaling applied to both features and target variable

This step improves model convergence and stability.

3.6 Feature Engineering

Feature engineering is performed to capture temporal and seasonal characteristics of demand.

- Temporal features: year, month, quarter
- Cyclical encoding: month_sin and month_cos
- Lag features: 1, 3, and 12 months
- Rolling averages: 3-month and 12-month trends

These features help the model learn both short-term fluctuations and long-term patterns.

3.7 Geographic Client Simulation (Non-IID Federation)

To simulate real-world distributed environments, routes are grouped into 8 clients using K-Means clustering based on geographic coordinates.

- Each client represents a transport operator
- Data distribution varies across clients (non-IID)
- Reflects real-world operational diversity

This setup introduces realistic challenges such as data heterogeneity and client imbalance.

3.8 Local Model Training

Each client trains a local model using its private dataset.

- Model used: SGDRegressor
- Training performed locally only
- Warm-start initialization from global model
- Iterative updates across training rounds

This ensures privacy-preserving model training.

3.9 Federated Aggregation

The central server aggregates local model updates using:

- **FedAvg:** Weighted averaging of client parameters
- **FedProx:** Adds a proximal term to handle non-IID data

The aggregation process improves global model performance iteratively.

3.10 Dynamic Operator Onboarding

The system supports dynamic participation of clients:

- New operators can join during training
- Existing clients can be removed if needed
- Automatic inclusion in subsequent training rounds

This ensures scalability and adaptability.

3.11 Insights Engine

The insights engine transforms model predictions into actionable analytics that support decision-making for transport operators and authorities. The key functionalities include:

- **Trend Analysis:** Evaluates year-on-year changes in ridership to identify patterns of growth, stability, or decline across routes.
- **Anomaly Detection:** Uses Z-score analysis to detect sudden deviations in demand, helping identify unusual events or disruptions.
- **Service Pattern Classification:** Classifies routes based on weekday and weekend demand characteristics into commuter-dominant, leisure-oriented, or balanced services.
- **Network Health Score:** Computes an aggregated performance metric that reflects overall system demand stability and operational efficiency.

3.12 Evaluation and Performance

The model is evaluated using the following metrics:

- Mean Absolute Error (MAE): 4,732

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- Error rate: approximately 2.7%
- Convergence achieved within 3–4 rounds

The results demonstrate that the proposed federated learning approach achieves high accuracy while maintaining data privacy.

Chapter 4

RESULTS AND DISCUSSIONS

This chapter presents the results obtained from the Privacy-Preserving Public Transport Demand Forecasting System and discusses its performance, adaptability, and operational capabilities. Experiments were conducted on real-world CTA ridership datasets, simulating multiple federated clients with heterogeneous data distributions.

4.1 Global Model Performance

The federated learning system was evaluated on held-out test data across all active clients. Key performance metrics are:

- **Final MAE:** 4,732 rides/route/month
- **Average monthly demand:** 176,000 rides
- **Approximate error rate:** 2.7%

These results demonstrate that the federated model predicts passenger demand accurately while preserving client privacy, highlighting the system's ability to generate actionable forecasts.

4.2 Federated Learning Convergence

The global model's convergence was monitored across 5 training rounds:

- Both **FedAvg** and **FedProx** aggregation algorithms achieved convergence by round 4.

- Round-by-round MAE reduction demonstrates iterative improvement, showing the system’s learning capability in a federated environment.
- FedProx shows a slightly higher initial MAE due to the proximal regularization term, stabilizing convergence for clients with diverse data.

4.3 Client-Level Performance

Local performance varies due to differences in route counts, data distribution, and demand patterns:

- MAE ranged from 2,281 (homogeneous, concentrated routes) to 7,043 (heterogeneous, mixed-use routes).
- Clients with smaller or more uniform datasets benefit more from the shared global model.
- Warm-starting from global parameters mitigates variability for smaller clients, improving local prediction reliability.

4.4 Dynamic Client Onboarding

The system supports dynamic onboarding of new operators without restarting federation:

- New clients can be added mid-training; the global model integrates their data in subsequent rounds.
- Temporary adjustments in global MAE reflect distributional shifts introduced by new operators.
- The model adapts seamlessly, demonstrating robustness to federation composition changes.

4.5 Insights Engine and Operational Intelligence

The three-layer insights engine transforms model outputs into actionable operational intelligence:

- **Trend analysis:** Routes classified as growing, stable, or declining using year-on-year comparisons.
- **Anomaly detection:** Sudden demand shocks flagged via Z-score analysis.
- **Service pattern classification:** Routes categorized as commuter-dominant, leisure/mixed, or balanced, aiding schedule optimization.

A composite **network health score** quantifies overall transit stability (e.g., 84/100), providing a single operationally meaningful indicator.

4.6 Discussion

- The system achieves accurate demand forecasting while maintaining strong privacy guarantees: raw data never leaves local operators.
- SGDRegressor is effective for federated learning: iterative, explainable, and compatible with small clients.
- FedAvg and FedProx produce similar final performance, with FedProx offering smoother convergence in heterogeneous scenarios.
- Dynamic onboarding confirms the system’s resilience: new operators can be integrated or removed without retraining from scratch.
- The insights engine converts model predictions into actionable guidance for transit authorities, demonstrating the system’s practical value beyond mere data analysis.

Chapter 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project presented a **Privacy-Preserving Public Transport Demand Forecasting System** based on Federated Learning (FL), enabling multiple transport operators to collaboratively train a global prediction model without sharing raw passenger data. The proposed system successfully addresses key challenges in real-world transport analytics, particularly data privacy, distributed data ownership, and non-IID data distributions across operators.

The system demonstrated strong predictive performance, achieving an approximate error rate of $\sim 2.7\%$, while maintaining efficient convergence within a few federated training rounds. The implementation of FedAvg and FedProx algorithms validated the effectiveness of federated optimization under heterogeneous data conditions. Additionally, the use of a lightweight SGDRegressor model ensured faster convergence, reduced communication overhead, and improved suitability for real-world deployment compared to computationally intensive deep learning models.

A key contribution of the system is the integration of an insights engine, which transforms predictions into actionable intelligence through trend analysis, anomaly detection, and service pattern classification. The system also supports dynamic client onboarding, demonstrating adaptability to changing participation in distributed environments. Overall, the project establishes that federated learning is a practical and scalable approach for collaborative demand forecasting in privacy-sensitive public transport systems, bridging the gap between predictive accuracy and real-world operational constraints.

5.2 Future Work

The proposed system can be further enhanced through the following improvements:

- **Personalised Federated Learning:** Fine-tuning the global model for individual clients to improve performance in highly heterogeneous environments.
- **Advanced Models:** Extending the system to support federated LSTM or Transformer-based models for capturing complex temporal patterns.
- **Enhanced Privacy Mechanisms:** Incorporating differential privacy or secure aggregation techniques for stronger data protection.
- **Scalable Deployment:** Implementing a fully distributed architecture with asynchronous communication and fault tolerance.
- **Real-Time Forecasting:** Enabling online federated learning for continuous and dynamic demand prediction.
- **Expanded Insights Engine:** Integrating additional analytics such as weather impact analysis and long-term forecasting.
- **Operational Integration:** Connecting predictions with scheduling, fleet management, and real-time decision-support systems.

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APPENDIX

5.3 Screenshots

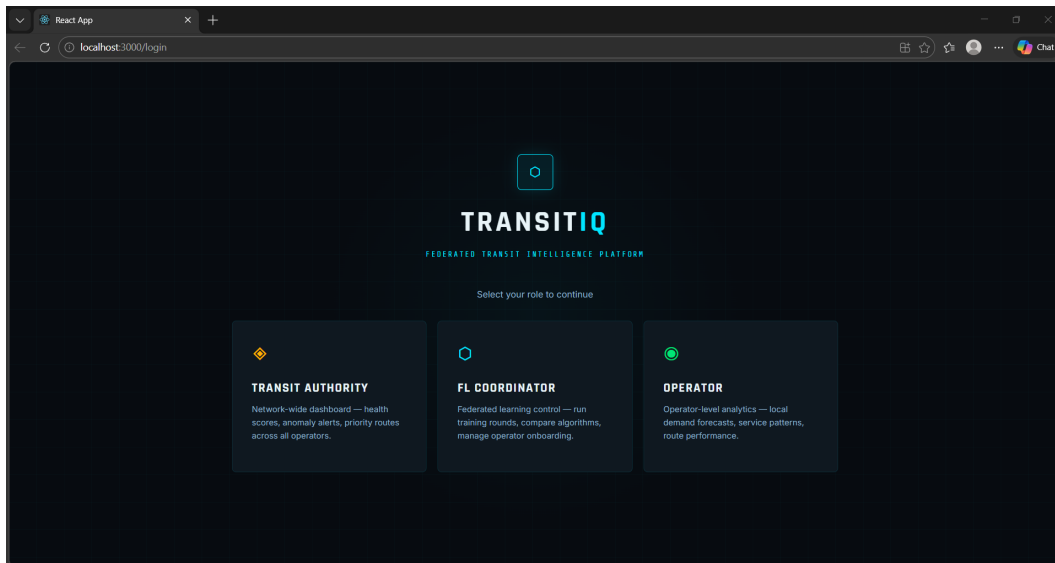


Figure 5.1: Homepage with Role Selection

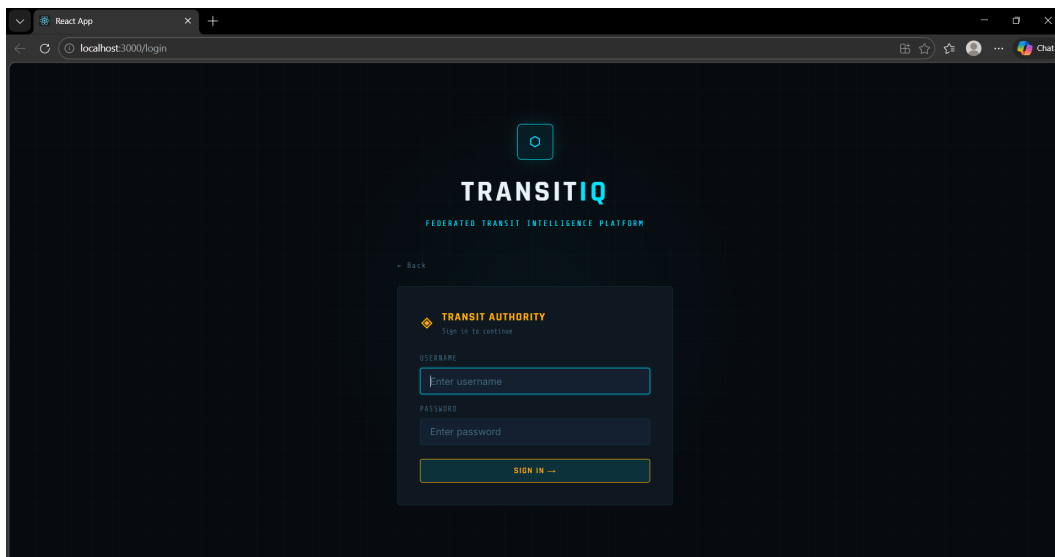


Figure 5.2: Transit Authority Login

Privacy-Preserving Public Transport Demand Forecasting System Using Federated Learning

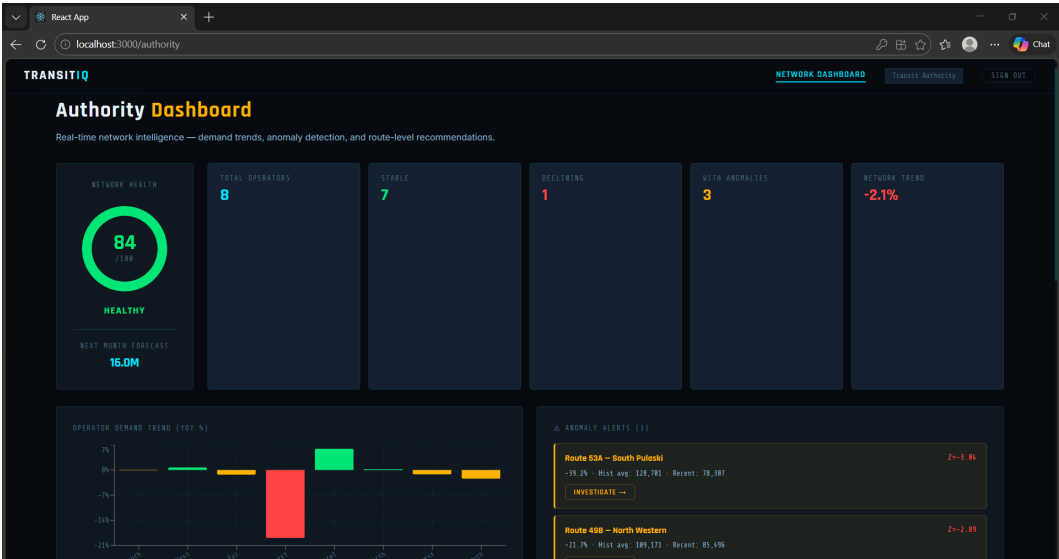


Figure 5.3: Transit Authority Dashboard Overview

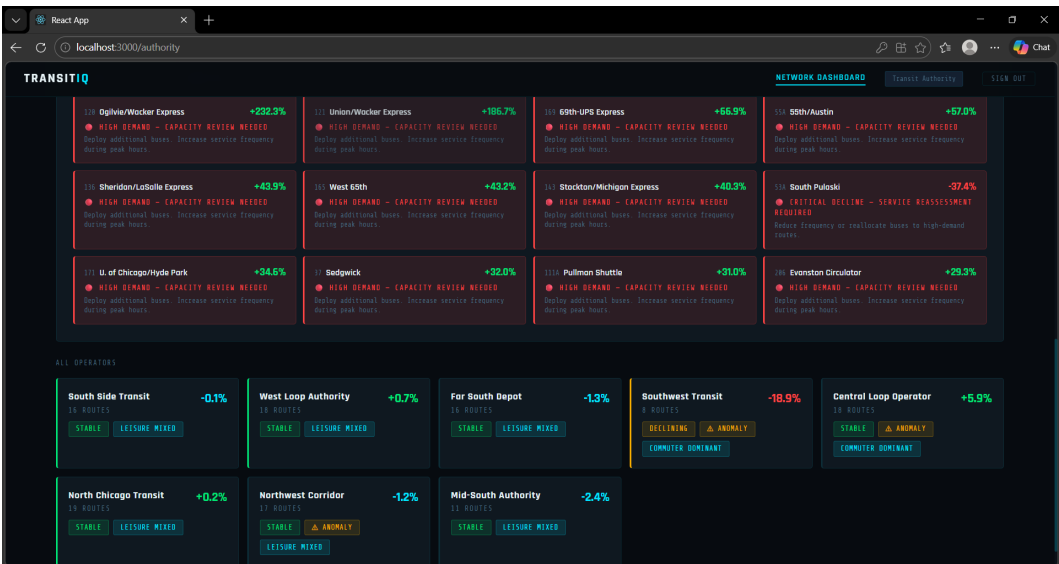


Figure 5.4: Authority Dashboard

Privacy-Preserving Public Transport Demand Forecasting System Using Federated Learning

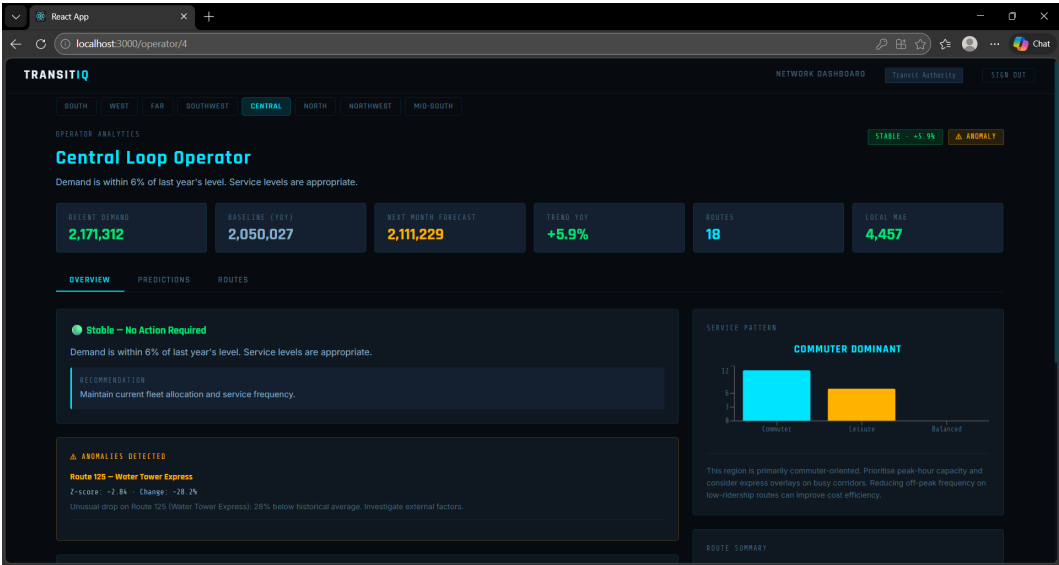


Figure 5.5: Operator-wise Analysis View from Transit Authority Dashboard

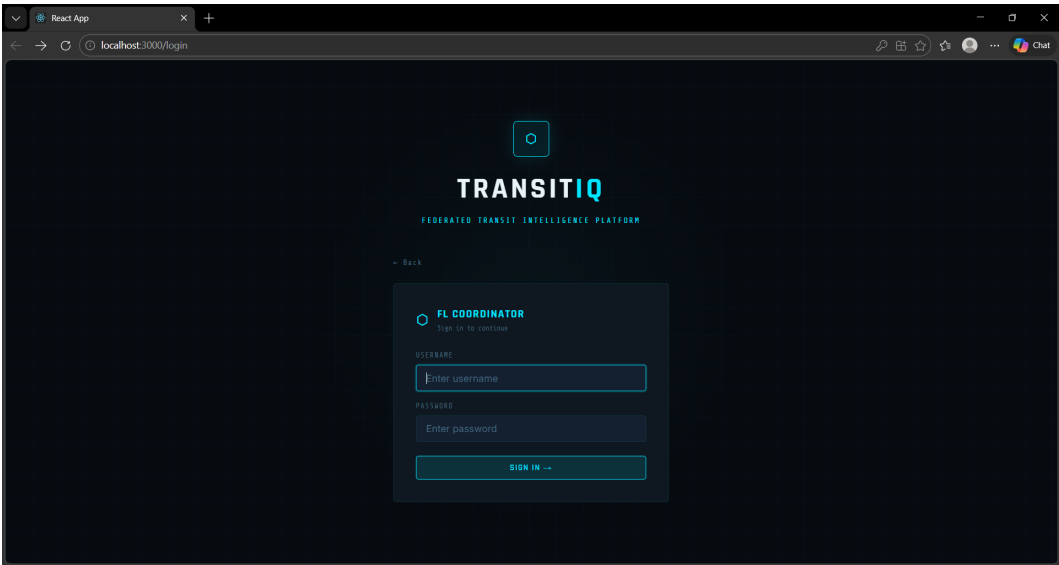


Figure 5.6: FL Coordinator Login

Privacy-Preserving Public Transport Demand Forecasting System Using Federated Learning

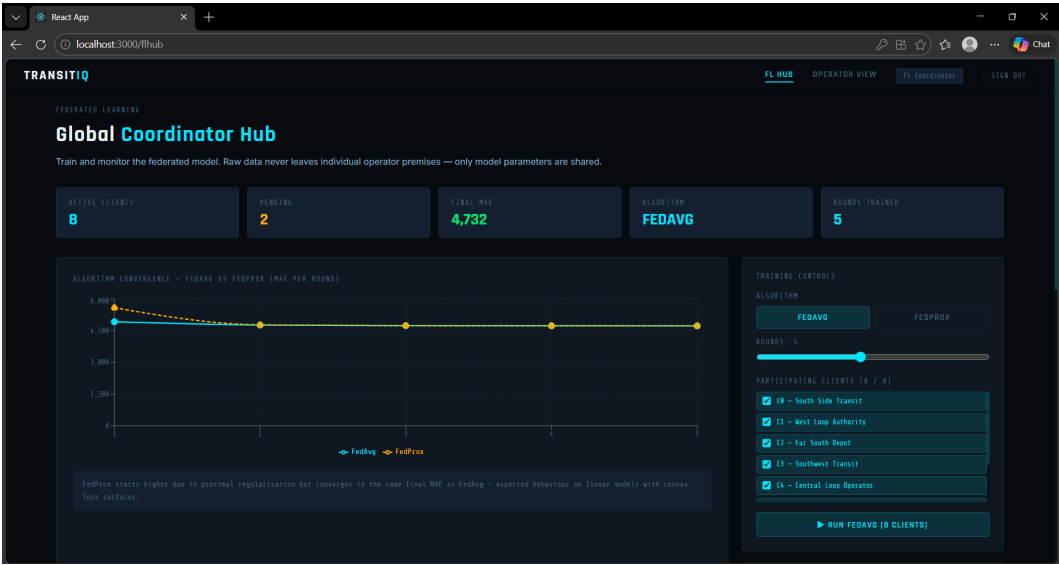


Figure 5.7: FL Coordniator Dashboard

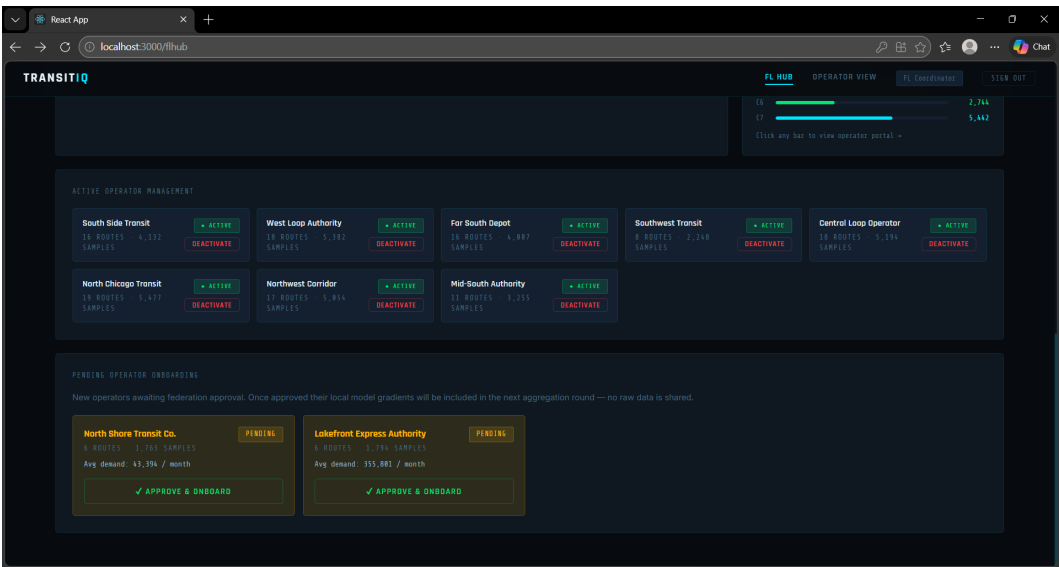


Figure 5.8: Active and Pending Clients in FL Coordinator Dashboard

Privacy-Preserving Public Transport Demand Forecasting System Using Federated Learning

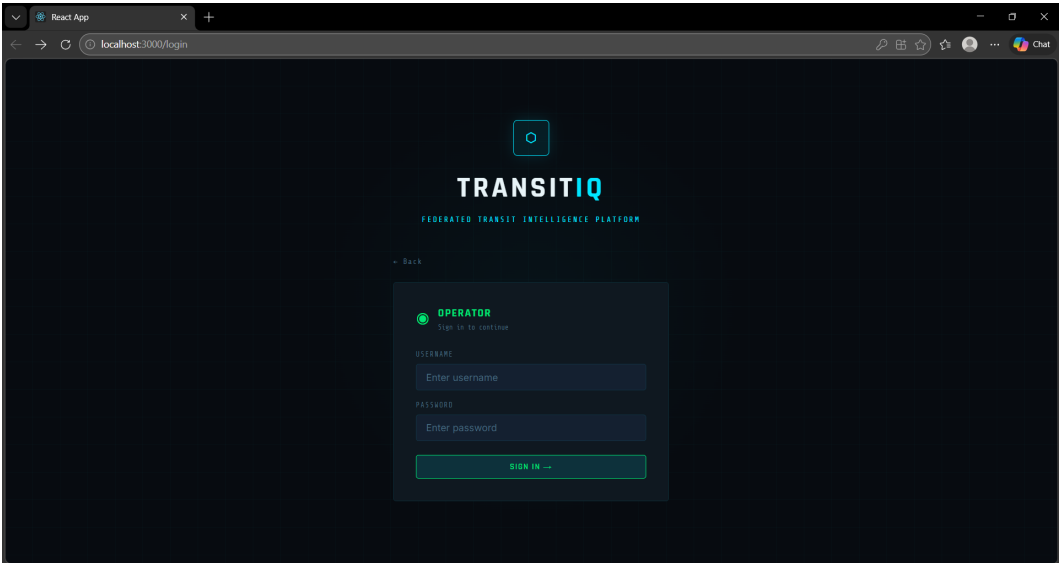


Figure 5.9: Operator Login

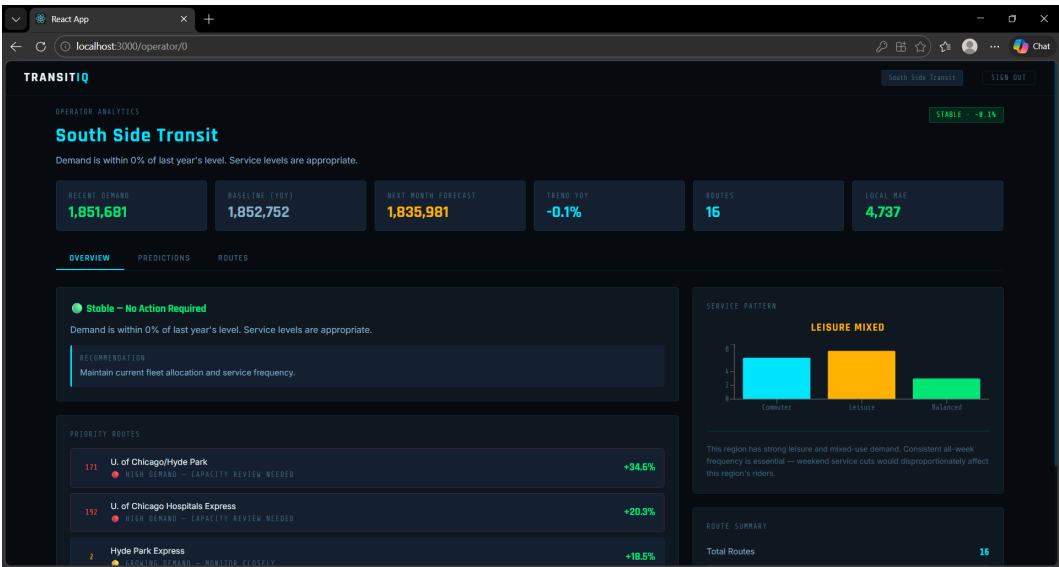


Figure 5.10: Operator Dashboard

Privacy-Preserving Public Transport Demand Forecasting System Using Federated Learning

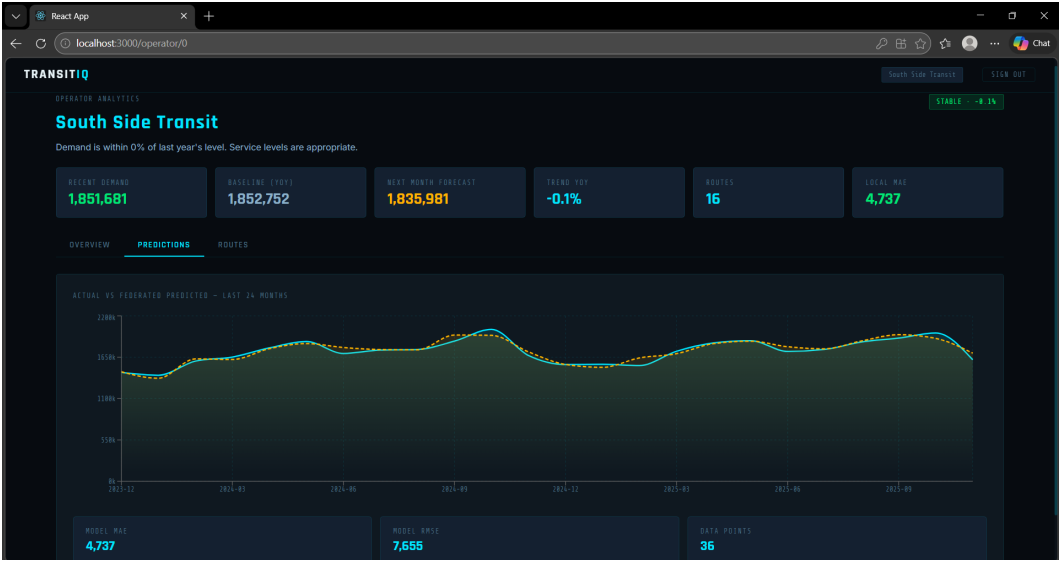


Figure 5.11: Actual vs Federated Predicted Graph in Last 24 Months

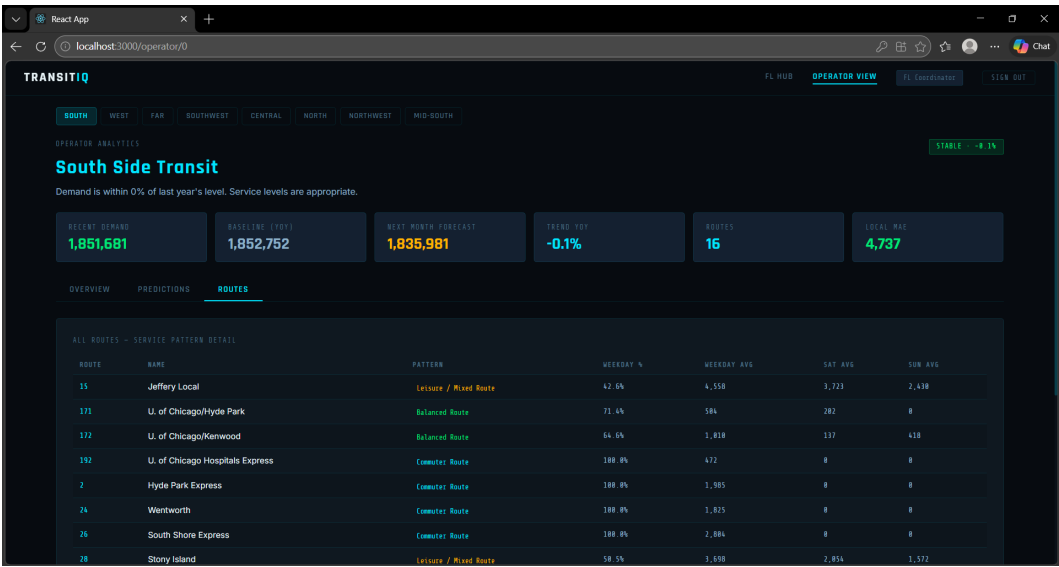


Figure 5.12: Service Pattern Classification of all routes